

Zen Nguyen

(672) 673-2613 | ttn54@sfu.ca | linkedin.com/in/zennguyen1305 | github.com/ttn54 | zennguyen.me

TECHNICAL SKILLS

- **Languages:** Python, Go, Java, C++, TypeScript, JavaScript
- **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC, Raft Consensus Algorithm, PostgreSQL, MongoDB, Pytest, JUnit
- **Infrastructure & DevOps:** AWS, Docker, Git, Linux
- **AI & Data:** RAG Architecture, Vector Databases, LangChain, OpenAI API, Gemini API, Pandas

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot)

Jan 2026 - Present

Personal Project · Python, LangChain, RAG Architecture, Gemini API

- Automated end-to-end job application workflows by architecting an autonomous AI Co-Pilot using Python and **LangChain**, resulting in uniquely tailored, ATS-optimized PDF resumes.
- Built a zero-hallucination **RAG (Retrieval-Augmented Generation)** engine using LangChain, Gemini APIs, and a FAISS Vector Store, delivering uniquely tailored, ATS-optimized PDF resumes for each job using only verified user data.
- Engineered a scraping pipeline using concurrent JobSpy scrapers across LinkedIn and Indeed, enabling the discovery, extraction, and parsing of real-time Software Engineering postings.
- Applied Test-Driven Development (TDD) with a 113-test suite using Pytest to validate all components including the scraper, AI tailor, PDF generator, and repository layer.

TitanStore (Distributed Raft Database)

Jan 2026 - Present

Personal Project · Go, Raft Consensus Algorithm, TCP Sockets, WAL

- Developed a **distributed key-value database** in Go from scratch, implementing the **Raft consensus protocol** for leader election and fault-tolerant log replication across a cluster of nodes.
- Engineered a crash-safe Write-Ahead Log (WAL) with atomic snapshots to guarantee data durability and enable node recovery after failures without data loss.
- Designed and implemented a high-throughput raw TCP server on port 6001, parsing a custom text protocol supporting SET, GET, and ERR NOT_LEADER operations with non-blocking I/O.
- Ensured data durability by implementing a crash-safe Write-Ahead Log (WAL) with atomic snapshots, guaranteeing zero data loss across all operations.

SFU Course Tracker (sfucourseplanner.me)

Nov 2025 - Jan 2026

Personal Project · Python, Docker, AWS, FastAPI

- Migrated full-stack platform from Azure to **AWS (EC2)**, implementing **Docker** container orchestration to optimize deployments and reduce infrastructure costs.
- Engineered a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, increasing data throughput by 10x while respecting server constraints.
- Designed and implemented a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST).

EDUCATION

Bachelor of Science, Computing Science

May 2025 - Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025